



3.4 Vectors (HL)

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A vector in 2D or 3D can be written as a column vector or using i, j, k , with magnitude $|v| = \sqrt{x^2 + y^2 + z^2}$. Vectors add/subtract componentwise and scale by a scalar; the displacement vector between points A and B is $\vec{AB} = \vec{B} - \vec{A}$ (position vectors). The scalar (dot) product $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$ gives the angle θ between two vectors, and $\vec{a} \cdot \vec{b} = 0$ tests for perpendicularity. A line through point with position vector $\vec{a}$, in direction $\vec{b}$, has vector equation $\vec{r} = \vec{a} + t\vec{b}$.

Worked example: follow the reasoning

Given $A(1, 2, 3)$ and $B(4, -1, 5)$, find (a) vector $\vec{AB}$, (b) $|\vec{AB}|$, (c) the vector equation of line AB.
Solution: (a) $\vec{AB} = \vec{B} - \vec{A} = (3, -3, 2)$. (b) $|\vec{AB}| = \sqrt{9 + 9 + 4} = \sqrt{22} \approx 4.69$. (c) $\vec{r} = (1, 2, 3) + t(3, -3, 2)$.

Quick check · multiple choice

If $\vec{a} = (3, 2, -1)$ and $\vec{b} = (1, -2, -1)$, find $\vec{a} \cdot \vec{b}$.

- A -3
- B 3
- C 0
- D 5

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Points $A(2, -1, 4)$, $B(5, 3, -2)$ and $C(0, 1, 2)$ are given. (a) Find vectors $\vec{AB}$ and $\vec{AC}$. [8]
(b) Find $\vec{AB} \cdot \vec{AC}$. (c) Hence find the angle between $\vec{AB}$ and $\vec{AC}$, to the nearest degree. (d) State what a value of zero in part (b) would indicate about $\vec{AB}$ and $\vec{AC}$.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $AB = (3, 4, -6)$, $AC = (-2, 2, -2)$. (b) $AB \cdot AC = 3(-2) + 4(2) + (-6)(-2) = -6 + 8 + 12 = 14$. (c) $|AB| = \sqrt{61} \approx 7.81$, $|AC| = \sqrt{12} \approx 3.46$; $\cos \theta = 14 / (7.81 \times 3.46) \approx 0.518$, so $\theta = \arccos(0.518) \approx 59^\circ$. (d) A dot product of zero would mean AB and AC are perpendicular.

Quick check answer

C. $a \cdot b = 3(1) + 2(-2) + (-1)(-1) = 3 - 4 + 1 = 0$, so a and b are perpendicular.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.